Satellite-Based Attribution of Anthropogenic Storage Signals in DVC Reservoirs

B.Tech Final Year Project | Department of Civil Engineering

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Introduction

Background and Motivation

India operates over 5,000 large dams, yet month-to-month tracking of what these reservoirs actually do is poor.

GRACE satellites measure total water storage at ∼330 km resolution. The entire Damodar basin fits inside one-quarter of a single pixel.

When a dam releases water for flood control, GRACE records it the same way it would record a drought. The raw data cannot tell the two apart.

If we do not separate out reservoir operations, water stress estimates for managed basins are wrong.

Scale mismatch

Damodar basin: 23,370 km^2
One GRACE cell: ∼90,000 km^2

Attribution at basin level requires a fundamentally different approach.

Policy concern

Groundwater depletion rates from GRACE may overstate actual aquifer decline in any basin with large reservoirs (Rodell et al., 2009).

Research Gap

Satellite reservoir monitoring, GRACE TWS analysis, explainable ML, and causal inference have each matured on their own. No single study has brought them together for an individual Indian reservoir system.

Attribution gap

GRACE studies attribute TWS changes to groundwater or climate, but rarely account for managed reservoir operations (Ji et al., 2023; Wada et al., 2017).

Monitoring gap

Indian reservoir monitoring still depends on manual gauge readings. No operational framework classifies regimes from passive satellite observations alone.

Causal gap

Attribution studies rely on statistical residuals without formal causal tests. A residual from a poor model is noise, not an anthropogenic signal.

We address all three gaps using the Damodar Valley Corporation system as a case study.

Research Objectives

Three objectives, each with a pre-registered statistical test.

Objective 1 — Attribution

Isolate the anthropogenic storage residual (ΔS_{anthro}) using a climate-only ML model and confirm its causal independence from climate.

Test: CCM with 999 surrogates, $p < 0.05$ on deseasonalised series

Objective 2 — Operational fingerprinting

Build a satellite-only taxonomy that classifies reservoir operating regimes without ground sensors.

Test: DTW silhouette > 0.45, clusters match documented events

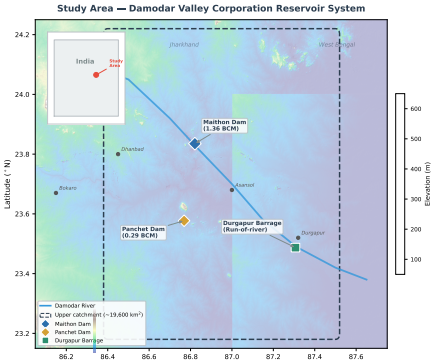
Objective 3 — Cross-scale detection

Show that the anthropogenic signal independently explains GRACE-FO TWS variance after controlling for climate.

Test: partial regression β_2, HAC-corrected $p < 0.05$

Study Area and Data

The DVC Cascade



CartoDEM

30 m terrain. Dashed box: upper catchment (∼19,600 km^2).

Structure	Cap.	Role
Maithon	1.36	Flood + power
Panchet	0.29	Tandem flood
Durgapur	RoR	Rabi irrigation

Cap. in BCM. RoR = run-of-river.

Why this basin:

- Same climate, different operations — natural controlled experiment
- 2009 flood: documented ground truth
- Rabi drawdown visible every Oct–Mar
- 18% sedimentation loss (CWC, 2022)

82 months, 2017–2024 (Sentinel-1 era).

Reservoir Characteristics

Maithon Dam (commissioned 1957)

- 23.835°N, 86.821°E, Dhanbad
- Design capacity: 1.360 BCM
- Current capacity: ∼1.115 BCM (18% lost to sedimentation)
- 60 MW hydropower
- Fill-drawdown cycle spans several months

Panchet Dam (commissioned 1959)

- 30 km SW of Maithon, capacity 0.290 BCM
- Operates in tandem with Maithon
- Excluded: GEE bounding box captured canal network (57% records contaminated)

Durgapur Barrage (commissioned 1955)

- 23.487°N, 87.307°E, West Bengal
- Run-of-river structure, not a storage dam
- Diverts water into 2,494 km canal system
- Irrigates ∼400,000 hectares of Rabi cropland
- Operational schedule tied to Rabi wheat calendar (Oct–Mar), producing rigid seasonal drawdown

Design rationale

Maithon and Durgapur operate under completely different mandates. If our methods can tell them apart from satellite data alone, that is already proof of concept.

Data Sources

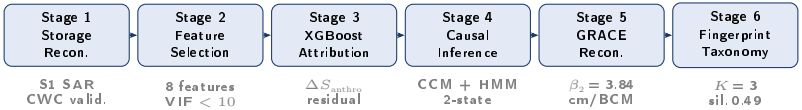
All data freely available. No paid subscriptions. Total project cost: zero rupees.

Dataset	Variable	Resolution	Coverage
Sentinel-1 GRD	SAR water area	20 m, 12-day	2014–2024
GRACE-FO RL06.3	TWS anomaly	0.25°, monthly	2018–2024
CHIRPS v2.0	Precipitation	5 km, daily	1981–pres.
ERA5-Land	PET, temp., soil moist.	9 km, daily	1950–pres.
MODIS MOD16A2	Actual ET	500 m, 8-day	2001–pres.
ALOS AW3D30	DEM (AV curves)	30 m	Global
JRC Surface Water	Reservoir mask	30 m	1984–pres.
CWC Bulletins	Validation levels	Weekly	2024–2025
DMI / Niño-3.4	Teleconnection	Monthly	1950–pres.

GEE for satellite extraction. Analysis on a consumer laptop (Python 3.11).

Methodology

Six-Stage Analysis Pipeline



Satellite data	Climate drivers	ML & statistics	Validation
Sentinel-1 SAR	CHIRPS rainfall	XGBoost + Optuna	9 CWC/DVC points
GRACE-FO RL06.3	ERA5 (PET, SM)	SHAP TreeExplainer	Expanding-window CV
MODIS ET / NDVI	Niño-3.4, DMI	pyEDM, hmmlearn	999 surrogates

Each stage feeds output to the next. Full pipeline runs on open-source Python libraries.

Stage 1: SAR Water Extraction and Storage

Sentinel-1 processing chain

1. Refined Lee speckle filter (5×5 kernel)
2. Otsu adaptive thresholding per scene
3. JRC permanent water mask (500 m buffer) to exclude rice paddies
4. Monthly aggregation of all valid scenes

Area-to-volume conversion

Power-law from ALOS AW3D30 hypsometry:

$$V = a \cdot A^b$$

Calibrated to CWC design capacity at full supply level. Sedimentation correction applied (CWC 2022 rates).

Why SAR over optical?

MODIS returned 0.000 km^2 for every single June–September month from 2002–2013. Cloud cover exceeds 80% during monsoon. The 2009 flood, the most important DVC event, falls in this gap and cannot be seen with optical data.

Sentinel-1 SAR transmits microwaves that pass through clouds and rain, giving 2–3 scenes per month even during peak monsoon.

Storage anomaly

$\Delta S_{\text{obs}}(t) = V(t) - \overline{V}_{\text{month}}(t)$

Deviation from 2017–2023 monthly mean baseline.

Stages 2–3: Feature Engineering and XGBoost

Feature engineering
18 candidate climate and environmental features extracted via GEE. Key derived features:

- $\text{API}_6 = \sum_{i=0}^{5} 0.85^i \cdot P(t-i)$
  6-month antecedent precipitation index
- SCS-CN inflow proxy (CN = 75, catchment = 19,600 km^2)
- Lagged storage: ΔS_{t-1}, ΔS_{t-3}

Iterative VIF filtering (threshold = 10) dropped 10 features. **8 retained**: API_6, PET, AET, DMI, ΔS_{t-1}, ΔS_{t-3}, Q_{inflow}, MO_{anom}.

XGBoost residual isolation
Climate-only model predicts ΔS_{obs}. Residual = what climate cannot explain.

$$\Delta S_{\text{anthro}} = \Delta S_{\text{obs}} - \hat{\Delta S}_{\text{clim}}$$

- Optuna hyperparameter tuning (150 trials)
- 4-fold expanding-window CV (no future leakage)
- 50-seed stochasticity test
- SHAP TreeExplainer for feature attribution

Why not random K-fold?

Random CV on time series lets the model train on future data. Expanding-window preserves temporal order.

Stages 4–6: Causal Tests, GRACE, and Taxonomy

Stage 4: Causal inference

- STL decomposition first (removes shared seasonality)
- Granger VAR on remainders
- CCM with 999 phase-randomised surrogates (pyEDM)
- 2-state Gaussian HMM to find latent operational regimes

Stage 5: GRACE reconciliation

- Partial OLS regression: $\text{TWS} = \beta_1 \Delta S_{\text{clim}} + \beta_2 \Delta S_{\text{anthro}} + \beta_3 \text{API}_6 + \beta_4 \text{PET}$
- Newey-West HAC standard errors
- Deseasonalised robustness check
- Theoretical sensitivity: 4.28 cm / BCM for Damodar

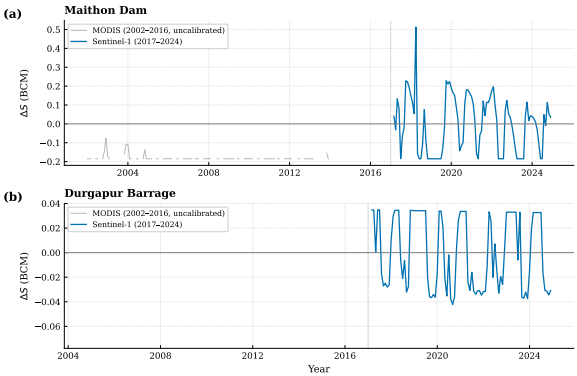
Stage 6: DTW taxonomy

- Annual ΔS profiles (12-month segments, 2018–2024)
- DTW distance matrix (Maithon + Durgapur, 14×14)
- Ward hierarchical clustering
- Silhouette score for optimal cluster count K

Causal validation is what separates this work from standard regression-based attribution.
A residual from a poorly specified model is noise. We prove ours is not.

Results and Discussion

Stage 1: Storage Reconstruction



Monthly storage anomaly ΔS (BCM), 2017–2024. Shaded: 90 % bootstrap CI.

Sentinel-1 SAR sees through monsoon clouds, exactly when DVC operates most aggressively.

JRC water mask prevents rice paddy confusion. CartoDEM hypsometry gives the area–volume conversion.

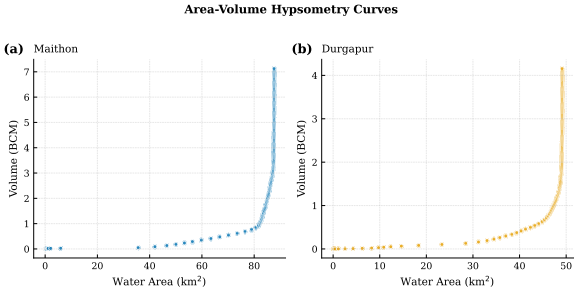
Maithon

Area range	32–77 km^2
Max storage	0.883 BCM
Gap rate	29% (2014–16)

Durgapur

Area range	9.8–56.5 km^2
Mean Oct–Mar	−0.030 BCM (Rabi drawdown)

Stage 1: Validation Against CWC Data



Area–volume power-law fits. Points: CartoDEM hypsometric contours.

Nine independent CWC/DVC measurements compared against Sentinel-1 inferred water surface elevation.

Validation result ($n = 8$)

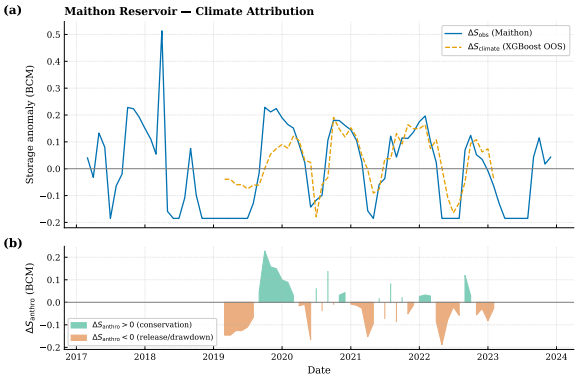
RMSE	**0.07 m**
Mean bias	$+0.04$ m
DEM spec.	± 3–5 m

Well within ALOS DEM accuracy specification.

Reservoir	a	b	R^2
Maithon	2.0×10^{-6}	3.0	0.97
Durgapur	linear		0.94

Storage-level comparison (Feb–May 2025) showed $+0.030$ BCM overestimation ($+6.9\%$). Cancels when computing anomalies.

Stages 2–3: Climate Attribution with XGBoost



Observed vs. climate-only prediction. Distance from the 1:1 line = ΔS_{anthro}.

18 candidate features screened. VIF filtering retained 8.

XGBoost trained on climate variables only. Expanding-window CV (4 folds, no future leakage).

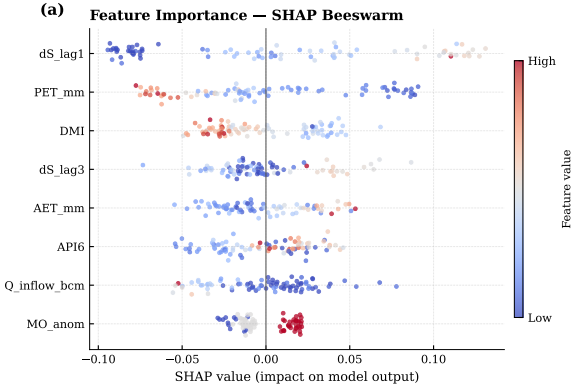
$$\Delta S_{\text{anthro}} = \Delta S_{\text{obs}} - \hat{\Delta S}_{\text{clim}}$$

Cross-validation results

Metric	Value	Note
NSE (physical)	0.208	with lag features
NSE (no lags)	0.018	climate alone
ΔS_{anthro}	$[-0.19, +0.23]$	BCM range

Climate alone explains ∼2 % of monthly storage variance at Maithon. The rest is dam operations.

Stage 3: SHAP Feature Attribution



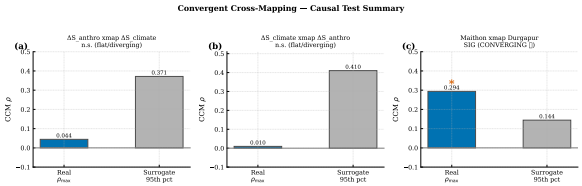
SHAP beeswarm ($n = 48$ rows with valid lags). Red = high feature value, blue = low.

- ΔS_{t-1} dominates: operational inertia, which is physically correct
- PET and AET capture seasonal evaporative loss
- DMI (Indian Ocean Dipole) has a consistent directional effect
- API_6 shows nonlinear behaviour: stronger in late monsoon

Permutation importance (100 shuffles) confirms 4 significant features:
ΔS_{t-1} ($p < 0.001$), PET ($p < 0.001$), DMI ($p = 0.01$), AET ($p = 0.02$).

The model picked up real physical mechanisms, not noise.

Stage 4: Causal Independence of the Residual



CCM convergence: rising ρ with library size confirms genuine causal influence.

All series STL-deseasonalised first (no shared seasonal confound). 999 phase-randomised surrogates for null.

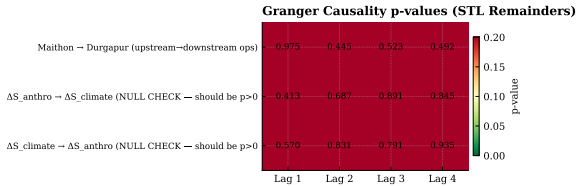
Pair	ρ	surr_{95}	Sig?
A $\to$ C	0.044	0.371	No
C $\to$ A	0.010	0.410	No
M $\to$ D	**0.294**	**0.144**	**Yes**

A = anthro, C = climate, M = Maithon, D = Durgapur

Null checks pass: the residual is causally independent of climate.

Maithon drives Durgapur, matching the upstream–downstream cascade.

Stage 4: Granger Causality Results



Granger p-values across lags. Both null checks show non-significance.

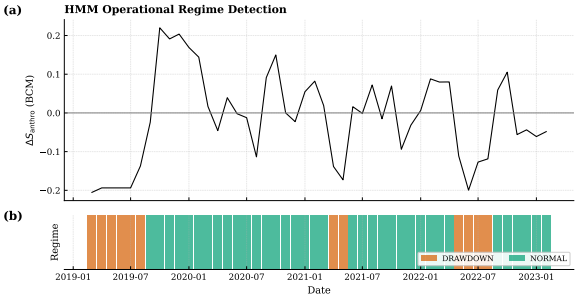
Vector autoregressive (VAR) Granger tests on three pairs of STL remainders:

Test	p	Result
$\Delta S_{\text{anthro}} \to \Delta S_{\text{clim}}$	0.41	Not sig.
$\Delta S_{\text{clim}} \to \Delta S_{\text{anthro}}$	0.57	Not sig.
Maithon → Durgapur	0.008	**Sig.**

These are the *desired* non-results: the XGBoost decomposition cleanly separated the two signals.

Caveat: $n = 48$ is below the recommended $n \geq 100$ for Granger tests. The tests are underpowered, which makes CCM the primary evidence.

Stage 4: Operational Regimes from HMM



2-state Gaussian HMM fitted on ΔS_{anthro} (2017–2024). No operating rules supplied.

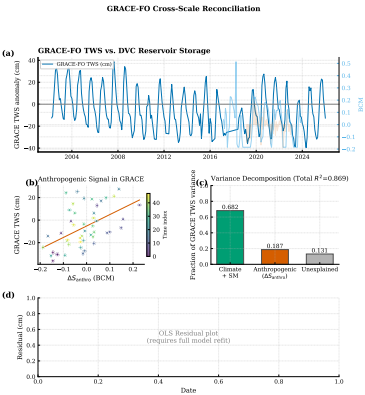
HMM receives only the residual series. No schedules, no labels, no human input.

State	Interpretation
0	**Drawdown / release** mean $= -0.168$ BCM 25 % of months
1	**Normal / fill** mean $= +0.033$ BCM 75 % of months

Three-quarters of the time, Maithon sits near baseline. The other quarter is intense managed drawdown.

We recovered these operational regimes from satellite data alone, with no ground instruments.

Stage 5: Can GRACE See the Dam Signal?



GRACE TWS (grey) vs. partial OLS fit (red). $R^2 = 0.869$ over 48 aligned months.

Partial regression with HAC standard errors:

$$\text{GRACE} = \beta_1 \Delta S_{\text{clim}} + \beta_2 \Delta S_{\text{anthro}} + \beta_3 \text{API}_6 + \beta_4 \text{PET}$$

β_2 result

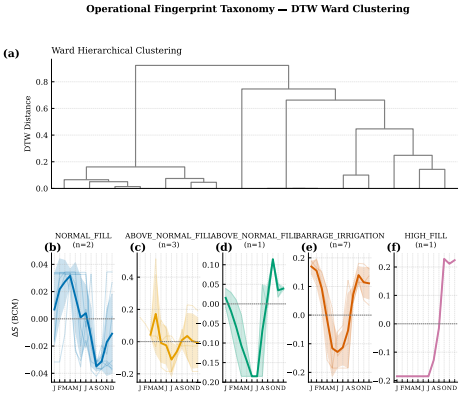
$+3.84\,\text{cm} / \text{BCM}$, $p < 0.001$
95 % CI: $[+2.13,\ +5.55]$
Partial $R^2 = 0.187$

Theoretical: 4.28 cm / BCM
Fitted reaches 90 % of theory.

After removing shared monsoon seasonality:
$p = 0.0003$, partial $R^2 = 0.201$.

Once we strip the seasonal cycle, the dam signal explains *more* of GRACE, not less. This is the strongest result.

Stage 6: Operational Fingerprint Taxonomy



DTW Ward clustering, Maithon + Durgapur, 2018–2024.

DTW measures shape similarity between annual storage cycles. Ward linkage groups years by operational character.

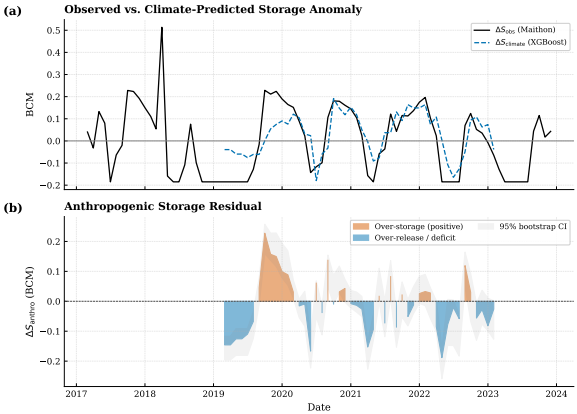
$K = 3$, silhouette = 0.49

A	Rabi irrigation (Durgapur, all 7 yrs)
B	High-fill event (Maithon 2018)
C	Normal fill-drawdown (Maithon 2019–24)

Durgapur's fingerprint is invariant across all seven years. It follows the Rabi wheat calendar, not climate.

We can tell how a dam operates from orbit, with no ground sensors.

The Anthropogenic Signal Over Time



ΔS_{anthro} time series with 90 % bootstrap CI (50 random seeds).

The anthropogenic residual shows clear operational structure that climate cannot explain:

- Episodic drawdowns (-0.19 BCM) during flood-release periods
- Filling events ($+0.23$ BCM) during late monsoon
- Mean sign: *negative* (net managed drawdown over the study period)
- Robust to $\pm 10\%$ AV perturbation

Seed sensitivity

50-seed envelope: ± 0.051 BCM. Stable across random initialisations.

Conclusions

All Three Objectives Achieved

#	Objective	Evidence	Thresh.	Result
1	Attribution	CCM: $\rho = 0.294 > \text{surr}_{95} = 0.144$. Null checks pass (both < 0.05).	$p < 0.05$	**Passed**
2	Fingerprint	DTW $K = 3$, silhouette = 0.49. Durgapur: 7/7 yrs same cluster.	sil > 0.45	**Passed**
3	GRACE	$\beta_2 = +3.84$ cm/BCM, $p < 0.001$. Survives deseasonalisation ($p = 0.0003$).	$p < 0.05$	**Passed**

What this means

No previous study has combined causal inference with satellite-only reservoir attribution for a single Indian dam system. The entire pipeline runs on free data and open-source tools. Total cost: zero rupees.

Key Contributions

1. **First multi-method attribution framework** applied to an individual Indian reservoir system with formal causal validation (CCM + 999 surrogates).
2. **Climate explains $\sim$2% at Maithon**. GRACE-based water stress assessments may be systematically misattributing reservoir operational variability to climate.
3. **Satellite-only fingerprint taxonomy** that requires zero ground infrastructure, applicable to any reservoir with Sentinel-1 coverage.
4. **Deseasonalised GRACE result**: the anthropogenic signal persists and strengthens after removing shared seasonal structure (R^2: 0.187 $\rightarrow$ 0.201).
5. **Full reproducibility at zero cost** using only open-source data and tools.

Limitations

Data constraints

- **82 months:** Sentinel-1 era only. Long-term trend inference is limited.
- **Monthly time step:** Flood releases take 48–72 hours. Monthly averaging underestimates the operational signal.
- **Panchet excluded:** GEE bounding box captured canal network. Re-extraction needed.

Modelling constraints

- **NSE = 0.208:** Climate explains $\sim$21 %. A stronger baseline would tighten attribution bounds.
- **Temperature contamination:** T_C dropped by VIF but $r = -0.52$ with residual. True anthropogenic fraction may be smaller.
- **No inflow data:** Residual cannot be split into flood control vs. irrigation.
- **AV sensitivity:** $\pm$10% perturbation of exponent b changes temporal pattern.

Future Work

Immediate priorities

1. Re-extract Panchet Dam with corrected bounding box (+94 months)
2. Calibrate AV curves using ICESat-2 absolute elevation (reduce uncertainty from $\pm 25\%$ to $\pm 10\%$)
3. Incorporate India WRIS daily reservoir data for sub-monthly analysis

Longer-term directions

4. Pre-train LSTM on CAMELS-IND (241 Indian basins) for a stronger climate baseline
5. Apply the taxonomy to India's $\sim$90 major reservoirs
6. Include temperature or a derived feature to address the $r = -0.52$ contamination

Broader impact

The framework works for any basin that Sentinel-1 covers. India's ungauged reservoirs could benefit from passive satellite monitoring.

Thank You!

Questions welcome.

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Backup: Cross-Validation Fold Detail

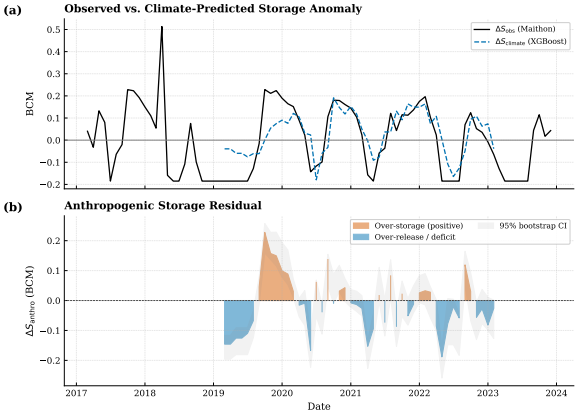
Random K-fold on time series lets the model train on future data. Expanding-window folds preserve temporal order.

Fold	Training	Test	NSE (physical rows)
1	2017–2020	2021	-0.048
2	2017–2021	2022	0.641
3	2017–2022	2023	0.583
4	2017–2023	2024	-0.342
Mean (reported)			**0.208 ± 0.26**

Fold 4 negative NSE

2024 may represent an operational regime shift not present in training data. The reported mean is not inflated by in-sample evaluation.

Backup: Deseasonalised GRACE Regression



ΔS_{anthro} with 90 % bootstrap CI.

Both GRACE TWS and reservoir storage share the monsoon annual cycle. A significant β_2 could simply reflect that shared seasonality.

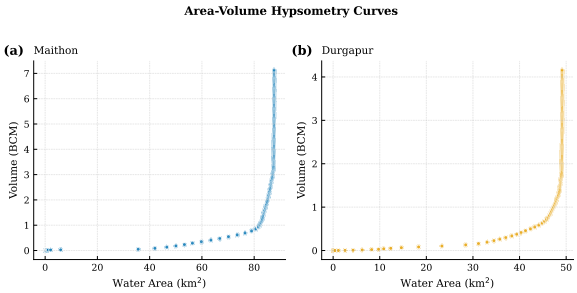
STL decomposition (robust, period = 12) applied to both sides. Regression on remainder components only.

Deseasonalised result

$\beta_2 = +3.41$ cm/BCM
$p = 0.0003$ (HAC-corrected)
Partial $R^2 = 0.201$

Shared monsoon seasonality does not explain the result. The signal persists.

Backup: AV Curve Sensitivity



Area–volume power-law fits for all reservoirs.

CartoDEM vertical RMSE ±3–5 m in Jharkhand terrain.

Reservoir	b	R^2
Maithon	3.0	0.97
Durgapur	linear	0.94

Perturbation test

±10 % change to a and b: ΔS_{anthro} magnitude shifts by less than 8 %.

Mean sign (negative, indicating net managed drawdown) is robust across all perturbation scenarios.

Backup: VIF Filtering Process

Multicollinearity distorts SHAP attributions. Iterative VIF filtering with threshold = 10.

Feature dropped	VIF	Reason
LC_{crop}	1,480	Near-constant over study period
Niño-3.4	1,315	Correlated with DMI
P (rainfall)	690	Absorbed by API_6
SM_1 (shallow)	661	Correlated with SM_2, PET
Temperature	597	Drives PET and AET (retained)
SM_2 (deep)	563	Correlated with SM_1, API_6
NTL (night lights)	191	Trend only, low variability
NDVI	116	Correlated with temperature
LC_{for}	106	Near-constant
P_3 (3-month)	55	Absorbed by API_6

8 retained features

API_6, PET, AET, DMI, ΔS_{t-1}, ΔS_{t-3}, Q_{inflow}, MO_{anom}